Lecture 4. Statistical Foundations

Exploratory and confirmatory analysis, samples and populations, sampling distributions, the bootstrap, outliers, models

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4.1 From description to statistical methods

Motivation. To go beyond the data in hand we must know how much the numbers would change with different data.

Statistical methods quantify patterns and their strength, find patterns too complex to see, and interpret data through models. **Exploratory** analysis checks integrity and generates hypotheses (transform–visualise–model); **confirmatory** analysis checks a model or hypothesis, for prediction or inference. Data used for exploration are not reused for confirmation.

The t-test for $H_0 : \mu = 0$ rejects when $|t| > 1.96$, $t = \sqrt{n}\,\bar{X}/s$; behind it lie

- a **model**: observations centred at an unknown μ with random error;
- a **fit**: centre and error estimated by $\bar{X}$ and s;
- **assumptions**: numeric, (approximately) normal observations;
- **uncertainty quantification**: the 95% confidence interval $\bar{X} \pm 1.96\, s/\sqrt{n}$;
- **strength of evidence**: the smaller the p-value, the stronger the evidence against the hypothesis.

Key idea: the data are a **sample** from a **population**; imagine the other samples that might have been drawn.

4.2 Samples and populations: a travel policy

Motivation. Knowing the whole population shows how much a small sample can mislead.

How early should a New York traveller's flight be scheduled to arrive in San Francisco so that delays rarely make them late? Treat the 2013 SFO flights as the population:

```
SF <- flights %>% filter(dest == "SFO", !is.na(arr_delay))
set.seed(101)
sf_25 <- SF %>% slice_sample(n = 25)
sf_25 %>% fivenum_tbl(arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1    25  0.12  29.8   -38   -23     -5    14   103
```

```
SF %>% fivenum_tbl(arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1 13173  2.67  47.7   -86   -23     -8    12  1007
```

(`set.seed()` makes `slice_sample()` reproducible.) The sample maximum is under two hours; the population has 1007 minutes. Accept a 2% risk, i.e. use the 98th percentile:

```
sf_25 %>% summarize(q98 = quantile(arr_delay, p = 0.98))
```

```
## # A tibble: 1 x 1
##     q98
##   <dbl>
## 1  67.5
```

```
SF %>% group_by(arr_delay < 90) %>% count() %>% mutate(pct = n / nrow(SF))
```

```
## # A tibble: 2 x 3
## # Groups:   arr_delay < 90 [2]
##   `arr_delay < 90`     n    pct
##   <lgl>            <int>  <dbl>
## 1 FALSE              640 0.0486
## 2 TRUE             12533 0.951
```

```
SF %>% summarize(q98 = quantile(arr_delay, p = 0.98))
```

```
## # A tibble: 1 x 1
##     q98
##   <dbl>
## 1   153
```

A 90-minute policy fails about 5% of the time; the population says about 150 minutes. The question is how good a sample estimate is.

4.3 Sampling distributions

Motivation. A statistic is random; its standard error measures its reliability.

A **statistic** summarises data (mean, sd, median); extremes such as the maximum are unreliable. Draw many samples from the population:

```
n <- 25
num_trials <- 500
sf_25_means <- 1:num_trials %>%
  map_dfr(~ SF %>% slice_sample(n = n) %>% summarize(mean_arr_delay = mean(arr_delay))) %>%
  mutate(n = n)
sf_25_means %>% fivenum_tbl(mean_arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  3.09  10.2 -20.1 -3.72   1.54     9  56.7
```

Vocabulary:

- the **sample size** n;
- the **sampling distribution**: the statistic over many trials;
- its **shape**: here somewhat right-skewed;
- the **standard error**: its standard deviation;
- the **95% confidence interval**: roughly mean ± 2 SE.

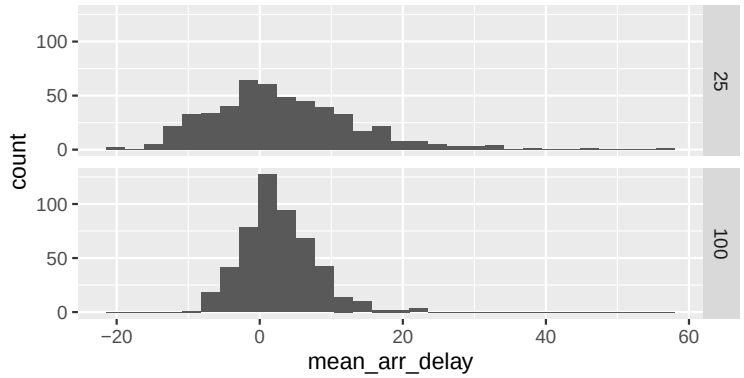
For $n = 25$ the means span about 75 minutes. With $n = 100$:

```
sf_100_means <- 1:num_trials %>%
  map_dfr(~ SF %>% slice_sample(n = 100) %>% summarize(mean_arr_delay = mean(arr_delay))) %>%
  mutate(n = 100)
bind_rows(sf_25_means, sf_100_means) %>%
  group_by(n) %>%
  summarize(mean = mean(mean_arr_delay), se = sd(mean_arr_delay))
```

```
## # A tibble: 2 x 3
##       n  mean    se
##   <dbl> <dbl> <dbl>
```

```
## 1    25  3.09 10.2
## 2   100  2.58  4.99
```

```
bind_rows(sf_25_means, sf_100_means) %>%
  ggplot(aes(mean_arr_delay)) + geom_histogram(bins = 30) + facet_grid(n ~ .)
```



Quadrupling n halves the standard error and makes the distribution more symmetric.

The $1/\sqrt{n}$ law [Supplement]

For a sample mean of i.i.d. observations, $\mathrm{Var}(\bar{X}) = \sigma^2/n$, so $\mathrm{SE} = \sigma/\sqrt{n}$: here $\sigma \approx 47.7$ gives 9.5 for $n = 25$ and 4.8 for $n = 100$, matching the simulation. (Without replacement multiply by $\sqrt{(N-n)/(N-1)} \approx 1$.) By the CLT the shape tends to normal, even for skewed delays.

4.4 The bootstrap

Motivation. With one sample, the bootstrap treats it as the population and resamples.

Resampling draws *with replacement* from the sample; duplication produces the variability. One sample of $n = 200$:

```
n <- 200
orig_sample <- SF %>% slice_sample(n = n, replace = FALSE)
sf_200_bs <- 1:num_trials %>%
  map_dfr(~ orig_sample %>% slice_sample(n = n, replace = TRUE) %>%
            summarize(mean_arr_delay = mean(arr_delay)))
sf_200_bs %>% fivenum_tbl(mean_arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  8.58  4.16 -1.94  5.47   8.42  11.3  23.7
```

```
sf_200_pop <- 1:num_trials %>%
  map_dfr(~ SF %>% slice_sample(n = n) %>% summarize(mean_arr_delay = mean(arr_delay)))
sf_200_pop %>% fivenum_tbl(mean_arr_delay)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  2.69  3.32 -6.41 0.391   2.65  4.86  18.0
```

The **bootstrap distribution** (first table) is centred at the sample mean, not the population mean, but its sd approximates the true SE (second table); for moderate n it approximates standard errors and quantiles (Efron and Tibshirani, 1993). The 98th percentile and its bootstrap reliability:

```
orig_sample %>% summarize(q98 = quantile(arr_delay, p = 0.98))
```

```
## # A tibble: 1 x 1
##     q98
##   <dbl>
## 1  209.
```

```
sf_200_bs_q98 <- 1:num_trials %>%
  map_dfr(~ orig_sample %>% slice_sample(n = n, replace = TRUE) %>%
            summarize(q98 = quantile(arr_delay, p = 0.98)))
sf_200_bs_q98 %>% fivenum_tbl(q98)
```

```
## # A tibble: 1 x 8
##       n  mean    sd   min   q25 median   q75   max
##   <int> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   500  201.  58.2  74.1  152.    208  251.   360
```

The SE is large: extreme quantiles need far more data than means.

4.5 Outliers

Motivation. Extreme delays suggest which conditions make long delays likely.

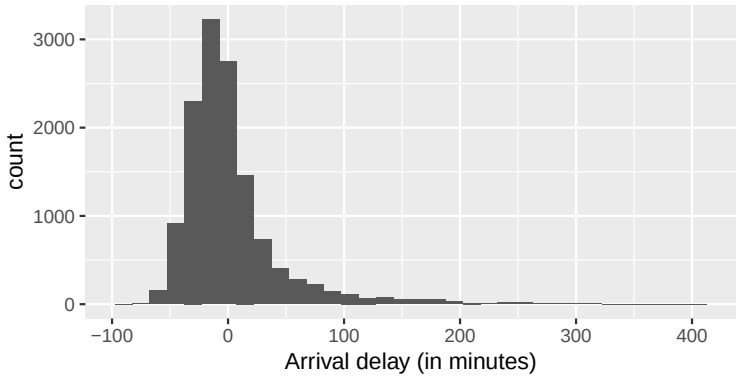
Delays of at least 420 minutes:

```
SF %>% filter(arr_delay >= 420) %>% select(month, day, dep_delay, arr_delay, carrier)
```

```
## # A tibble: 7 x 5
##   month   day dep_delay arr_delay carrier
##   <int> <int>     <dbl>     <dbl> <chr>
## 1    12     7       374       422 UA
## 2     7     6       589       561 DL
## 3     7     7       629       676 VX
## 4     7     7       653       632 VX
## 5     7    10       453       445 B6
## 6     7    10       432       433 VX
## # i 1 more row
```

Without them, most flights are on time or under an hour late.

```
SF %>% filter(arr_delay < 420) %>%
  ggplot(aes(arr_delay)) + geom_histogram(binwidth = 15) +
  labs(x = "Arrival delay (in minutes)")
```



Long delays (over an hour) by month and carrier:

```
SF %>% mutate(long_delay = arr_delay > 60) %>%
  count(month, long_delay) %>%
  pivot_wider(names_from = month, values_from = n)
```

```
## # A tibble: 2 x 13
##   long_delay   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <lgl>      <int> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
## 1 FALSE        856   741   812   993  1128   980   966  1159  1124  1177  1107  1093
## 2 TRUE          29    21    61   112    65   209   226    96    65    36    51    66
```

```
SF %>% mutate(long_delay = arr_delay > 60) %>%
  count(carrier, long_delay) %>%
  pivot_wider(names_from = carrier, values_from = n)
```

```
## # A tibble: 2 x 6
##   long_delay    AA    B6    DL    UA    VX
##   <lgl>      <int> <int> <int> <int> <int>
## 1 FALSE       1250   934  1757  6236  1959
## 2 TRUE         148    86    91   492   220
```

June and July are bad; Delta (DL) does well: arrive extra early in June–July, consider Delta.

4.6 Statistical models

Motivation. A model relates the response to several variables at once, separating signal from noise.

"Essentially, all models are wrong, but some are useful" (G. E. P. Box). We build models

1. to **understand** (which factors matter?), with simple interpretable models;
2. to **predict** unseen data, possibly with black boxes.

With response Y and predictors $X = (X_1, \ldots, X_p)$, we assume

$$Y = f(X) + \epsilon,$$

f unknown, $\mathbb{E}\epsilon = 0$. Estimate f for **prediction** $\hat{Y} = \hat{f}(X)$ or **inference** (which predictors matter, how, is linear enough).

Proposition 4.1 (reducible and irreducible error)

If $\hat{f}$ and X are fixed and $\mathbb{E}\epsilon = 0$, then

$$\mathbb{E}[(Y - \hat{Y})^2] = [f(X) - \hat{f}(X)]^2 + \mathrm{Var}(\epsilon).$$

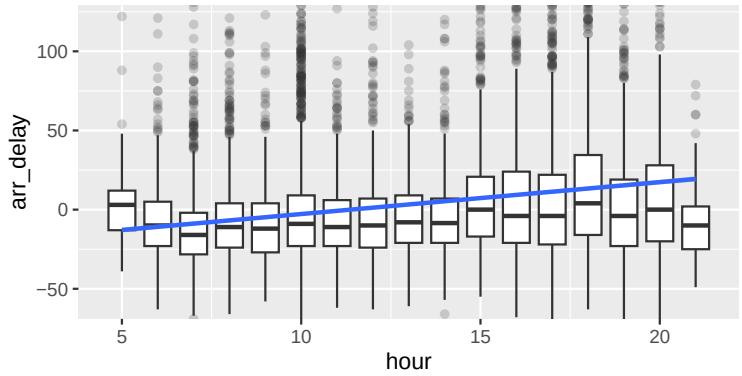
Proof [Supplement]. $Y - \hat{Y} = \{f(X) - \hat{f}(X)\} + \epsilon$, where the first term is a constant c. Hence $\mathbb{E}(Y - \hat{Y})^2 = c^2 + 2c\,\mathbb{E}\epsilon + \mathbb{E}\epsilon^2 = c^2 + \mathrm{Var}(\epsilon)$. ■

The first term (**reducible error**) shrinks with a better $\hat{f}$; the second (**irreducible error**) cannot.

Parametric models specify $f(X;\theta)$ with finitely many parameters, e.g. $Y \approx \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$; **nonparametric** models such as k-nearest neighbours assume no form for f. Flexibility costs interpretability.

Example. Do delays cascade through the day (SFO departures 5 am–9 pm)?

```
SF %>% ggplot(aes(hour, arr_delay)) +
  geom_boxplot(aes(group = hour), outlier.alpha = 0.2) +
  geom_smooth(method = "lm", formula = y ~ x) +
  coord_cartesian(ylim = c(-60, 120))
```




```
mod1 <- lm(arr_delay ~ hour, data = SF)
broom::tidy(mod1)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)   -22.9     1.23       -18.6 2.88e- 76
## 2 hour            2.01    0.0915      22.0 1.78e-105
```

Mean delay rises about 2 minutes per hour; the tiny p-value rules out no relationship.

4.7 Selected exercises

Motivation. Exact counterparts of the lecture's simulations.

Exercise 4.1 (what a resample contains)

A bootstrap resample draws n times with replacement from $x_1, \dots, x_n$.
(a) Show that a fixed observation x_i is absent from the resample with probability $(1-1/n)^n$, and that this tends to $e^{-1} \approx 0.368$.
(b) Show that the expected number of distinct original observations in a resample is $n\{1-(1-1/n)^n\} \approx 0.632\,n$.

Solution [Supplement].

(a) Each of n independent draws misses i with probability $1-1/n$, so $P(x_i \text{ absent}) = (1-1/n)^n$. With $t = 1/n \to 0$, $n\log(1-1/n) = \log(1-t)/t \to -1$ (derivative of $\log(1-t)$ at 0), so the limit is e^{-1} (0.367 for $n = 200$).

(b) With $I_i = 1\{x_i \text{ appears}\}$, by linearity and (a), $\mathbb{E}\sum_i I_i = \sum_i P(I_i = 1) = n\{1-(1-1/n)^n\} \approx (1-e^{-1})n = 0.632\,n$. ■

Exercise 4.2 (bootstrap standard error of a mean)

Condition on the data and let $\bar{x}^*$ be the mean of a bootstrap resample. Show that $\mathbb{E}^*\bar{x}^* = \bar{x}$ and $\mathrm{Var}^*\,\bar{x}^* = \hat{\sigma}^2/n$ with $\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2$, where $\mathbb{E}^*$, Var^* refer to resampling only. Compare with the usual estimate $s/\sqrt{n}$ of the standard error.

Solution [Supplement].

Step 1. The resampled values $x_1^*, \dots, x_n^*$ are i.i.d., each uniform on the multiset $\{x_1, \dots, x_n\}$. Hence $\mathbb{E}^* x_j^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\mathrm{Var}^*\, x_j^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$.

Step 2. By linearity, $\mathbb{E}^*\bar{x}^* = \bar{x}$ (centred at the sample mean).

Step 3. By independence, $\mathrm{Var}^* \bar{x}^* = \frac{1}{n^2}\sum_j \mathrm{Var}^* x_j^* = \hat{\sigma}^2/n$.

Step 4. Since $\hat{\sigma}^2 = \frac{n-1}{n}s^2$, the bootstrap SE is $\sqrt{(n-1)/n}\ s/\sqrt{n}$, i.e. $s/\sqrt{n}$ times 0.9975 for $n = 200$; simulation adds Monte Carlo error. ■

Exercise 4.3 (the finite-population correction)

A population has values $v_1, \dots, v_N$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum_j (v_j - \mu)^2$. A simple random sample $X_1, \dots, X_n$ is drawn without replacement ($n \le N$). Prove that

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

Solution [Supplement].

Step 1. By symmetry X_i is uniform on the population ($\mathbb{E}X_i = \mu$, $\mathrm{Var}\, X_i = \sigma^2$), and for $i \neq j$, (X_i, X_j) is uniform on ordered pairs of distinct units, so $c = \mathrm{Cov}(X_i, X_j)$ does not depend on i, j, n.

Step 2. Find c from the full census $n = N$: then $\sum_{i=1}^N X_i = \sum_j v_j$ is constant, so

$$0 = \mathrm{Var}\Big(\sum_{i=1}^N X_i\Big) = N\sigma^2 + N(N-1)c, \qquad c = -\frac{\sigma^2}{N-1}.$$

Step 3. For general n,

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\{n\sigma^2 + n(n-1)c\} = \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1}.$$

The factor is ≈ 1 for $n \ll N$ and 0 for a census. ■

Exercise 4.4 (how large a sample)

For the SFO arrival delays, $\sigma \approx 47.7$ minutes. How many flights are needed for the 95% confidence interval $\bar{X} \pm 1.96\,\sigma/\sqrt{n}$ for the mean delay to have half-width at most 5 minutes? By what factor must n grow to halve the half-width again?

Solution [Supplement]. $1.96\,\sigma/\sqrt{n} \le 5 \iff n \ge (1.96 \cdot 47.7/5)^2 = 18.698^2 = 349.6$, so $n = 350$. Half-width $\propto 1/\sqrt{n}$, so halving it needs $4n = 1400$. ■